

Worksheet 3: Inflorescence & Flower Architecture

NCERT Class 11 Biology — Chapter 5: Morphology of Flowering Plants

Student Name: _____	Roll No.: _____	Section: _____
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Section A: Multiple Choice Questions (Select the single best answer)

1. A flower is a modified shoot because: [_____]

- A) The shoot apical meristem transforms into floral meristem and internodes do not elongate
- B) It develops directly from the embryonic radicle
- C) It bears unicellular root hairs and root cap
- D) The floral axis undergoes infinite elongation without nodes

2. In a cymose inflorescence, the main floral axis: [_____]

- A) Continues to grow indefinitely in acropetal succession
- B) Terminates in a flower, showing determinate growth with basipetal succession
- C) Never bears bracts or pedicels
- D) Opens flowers centripetally from periphery to center

3. Which of the following sets of plants possess zygomorphic (bilaterally symmetrical) flowers? [_____]

- A) Mustard, Datura, Chilli
- B) Pea, Gulmohur, Bean, Cassia
- C) Canna, Mustard, Datura
- D) Brinjal, Plum, Rose

4. Flowers with a superior ovary, where the gynoecium occupies the highest point on the thalamus, are found in: [_____]

- A) Plum, Rose, Peach
- B) Guava, Cucumber, Ray florets of Sunflower
- C) Mustard, China rose, Brinjal
- D) Apple, Pear, Peach

Section B: Fill in the Blanks & Classifications

5. A flower that cannot be divided into two equal halves by any vertical radial plane is called _____ (e.g., Canna).

6. In perigynous flowers, the gynoecium is located at the center and other parts are on the rim of the thalamus at the same level; the ovary is termed _____.

7. When calyx and corolla are undifferentiated, individual members of the perianth are designated as _____ (e.g., Lily).

Section C: Comparative Inflorescence & Ovary Table

Flower Insertion Type	Thalamus & Floral Part Position	Ovary Status	NCERT Standard Examples
Hypogynous	Gynoecium highest; calyx/corolla/androecium below	Superior (G)	_____
Perigynous	Gynoecium in center; other whorls at rim same level	Half-inferior	_____
Epigynous	Thalamus encloses ovary completely & fuses; parts above	Inferior (G)	_____

Section D: Short Answer & Analytical Questions

8. Contrast Racemose and Cymose inflorescence on the basis of: (a) Axis growth, (b) Flower arrangement succession, and (c) Flower opening sequence.

9. Define Actinomorphic vs. Zygomorphic floral symmetry with diagnostic criteria and two examples each.

Lecturer Signature: _____